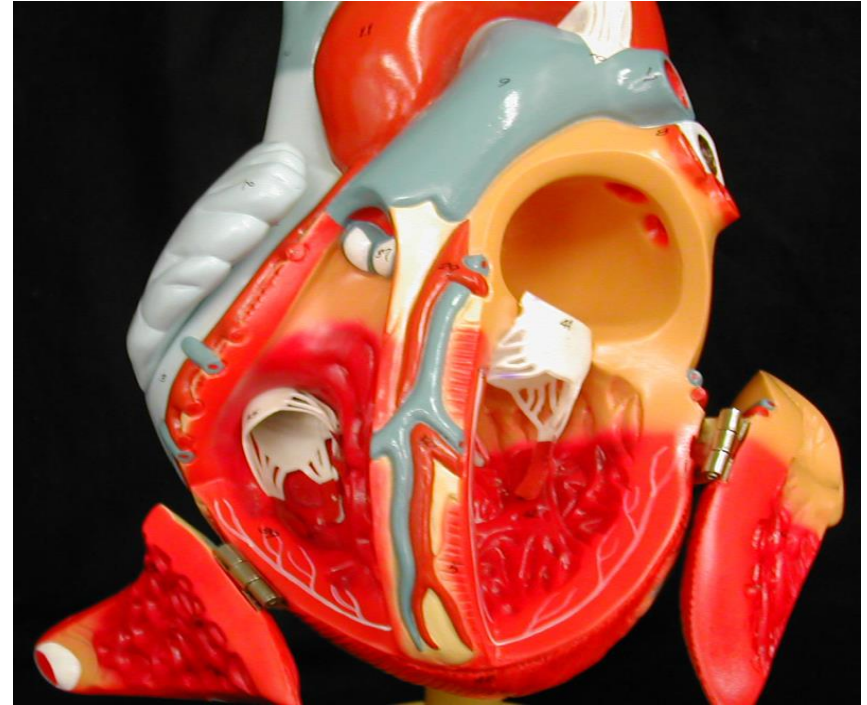


Cardiovascular System



William Harvey



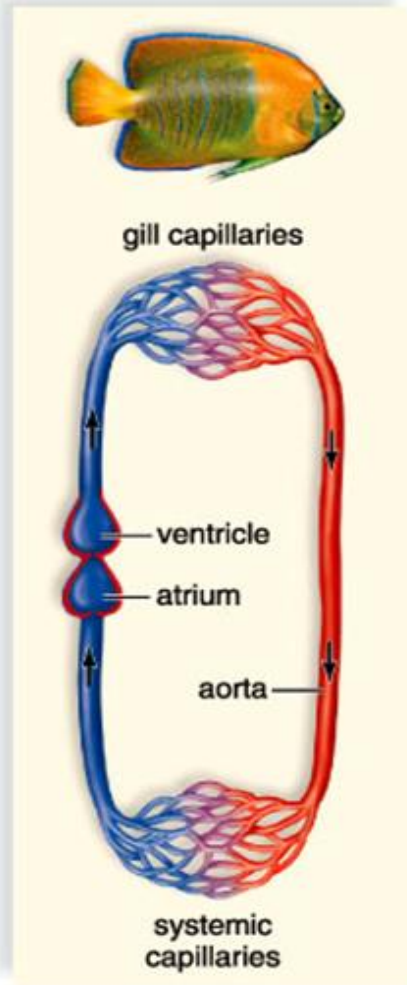
- Was an English physician.
 - He was the first to describe completely the function of heart in detail.
 - Was the first to recognize the full circulation of the blood in the human body.

Types of Circulation

- Blood Circulation is an evolutionary consequence of body size.
- Isolated small cells and small organisms with no circulatory system meet metabolic needs by simple processes of diffusion and convection of solutes.
- Animals that have a circulatory system have one of two kinds:
 - Open: fluid is circulated through an open body chamber.
 - Closed: fluid is circulated through blood vessels.
- In lower animals and vertebrates the circulation may be:
 - ❑ Open or closed circulation
 - ❑ Single or double circulation

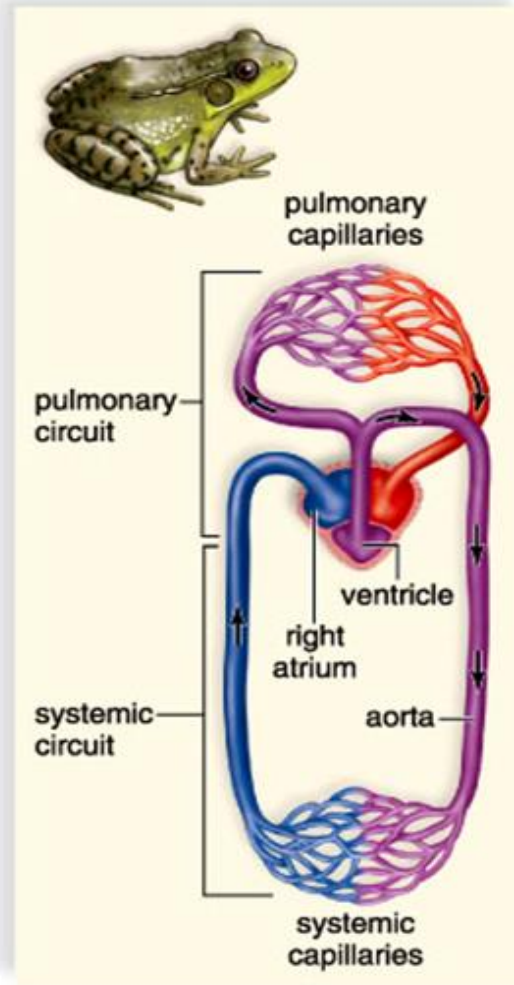
Comparison of Circulatory Circuits

single

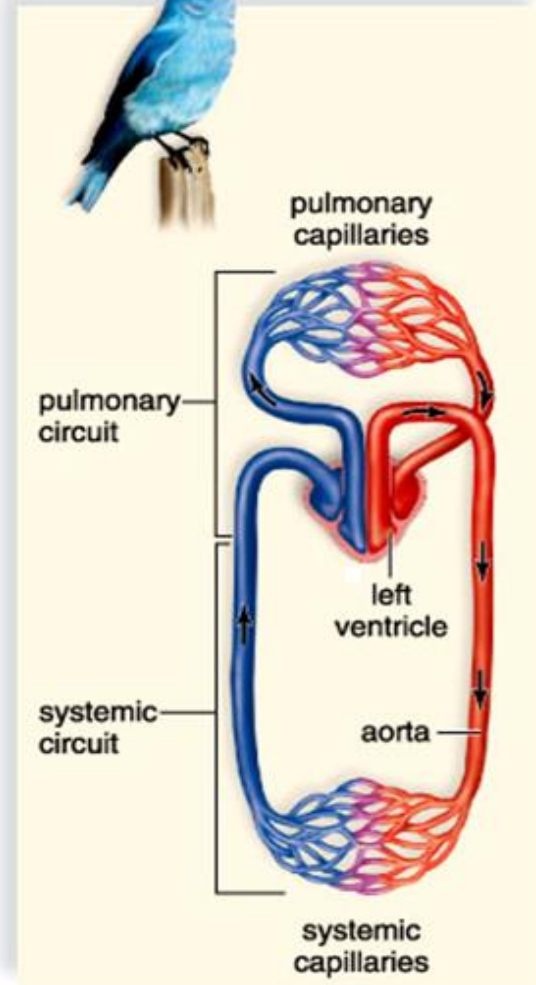


a. Fishes

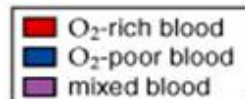
Double



b. Amphibians and most reptiles

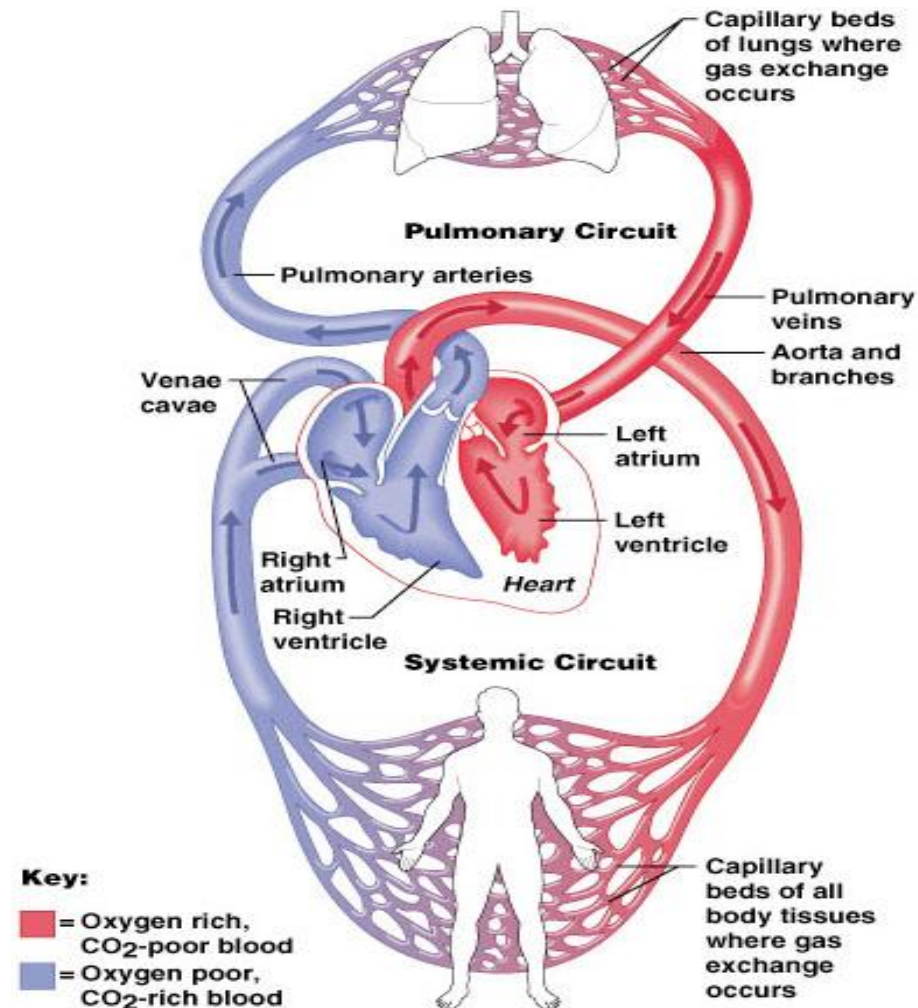


c. Some reptiles; birds and mammals



Double Circulations

- Human beings have a closed and double circulatory system.
- The heart pumps blood into two closed circuits with each beat.
 - Systemic circulation and pulmonary circulation.
- The output of one becomes the input of the other.



Portal Circulation

- It is a part of systemic circulation, which has the following characteristics.
- The blood passes through two sets of capillaries before draining into a systemic vein.
- The vein draining from the first capillary network is known as *portal vein* which branches like an artery to form the second set of capillaries or sinusoids.
 - Examples: *hepatic portal circulation, hypothalamo hypophyseal portal circulation and renal portal circulation.*

Introduction to CVS

- The cardiovascular system is a fluid transport system for the movement of blood throughout the body.
- Functional components of the cardiovascular system:
 - Heart
 - Blood Vessels
 - Blood

What is Heart

- It is the central pumping organ.
- For blood to reach body cells and exchange materials with them, it must be pumped continuously by the heart through the blood vessels.
- The heart beats about 100,000 times every day,
- 35 million beats in a year, and approximately 2.5 billion times in an average lifetime.
- Each day the heart pumps about 8000-14,000 liters of blood.

Introduction to CVS

- The left side of the heart pumps blood through an estimated 120,000 km (75,000 mil) of blood vessels.
- The right side of the heart pumps blood to the lungs, enabling blood to pick up oxygen and unload carbon dioxide.
- **Arteries**, or efferent vessels, carry blood away from the heart, and **veins**, or afferent vessels, return blood to the heart.
- Microscopic thin-walled vessels called **capillaries** interconnect the smallest arteries and the smallest veins.
 - Their thin walls permit the exchange of nutrients, gases, and waste products between the blood and surrounding tissues.

General Functions of the CVS

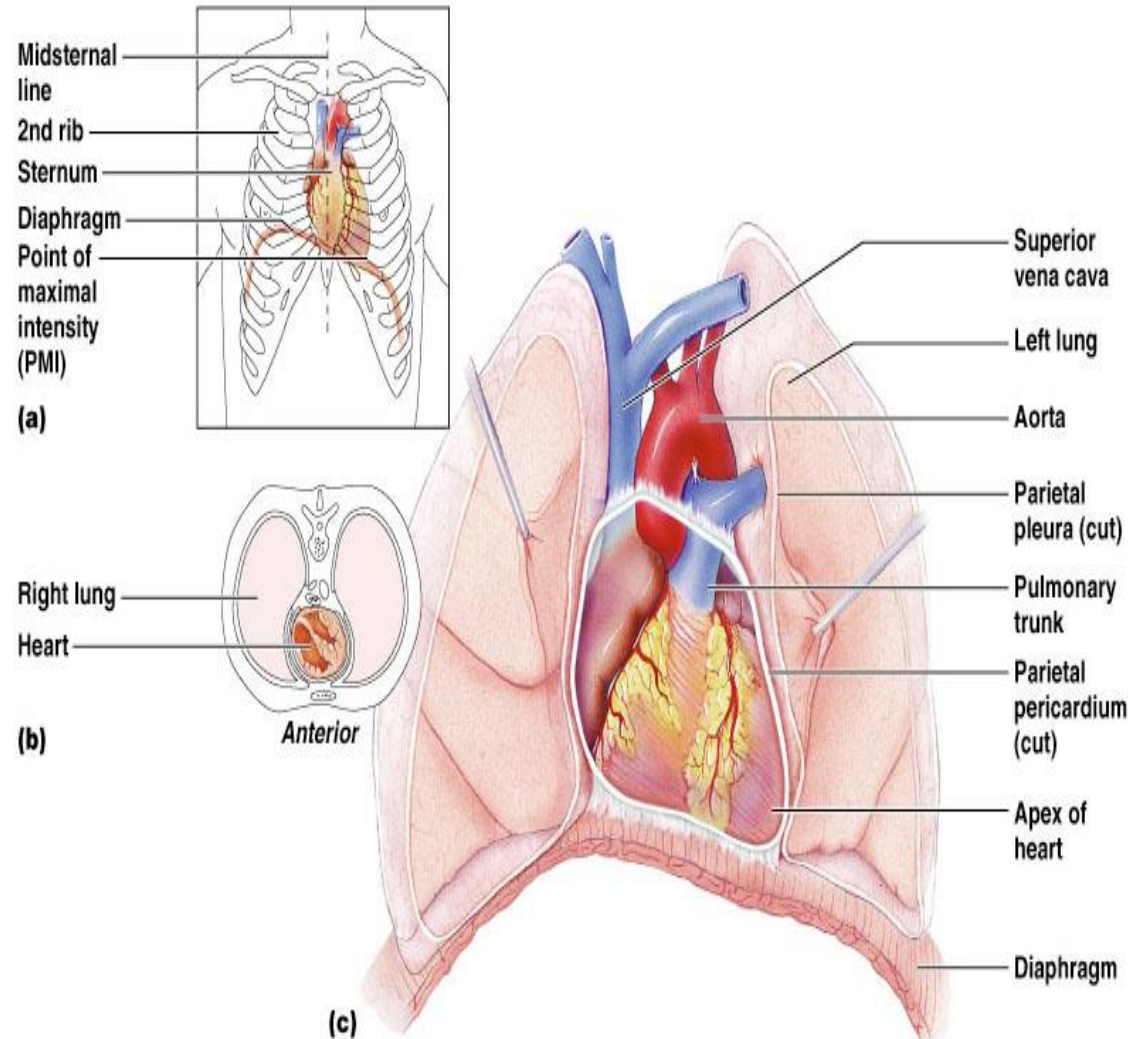
1. Convective (mass movement of fluid caused by pressure gradient) transport of O₂, nutrients, water, hormones, electrolytes, and drugs
2. Rapid washout of metabolic wastes
3. Control function relating to distribution of hormones to tissues and secretion of some hormones like ANP
4. Contribution to regulation of temperature and blood flow
5. Vital role in reproduction-hydraulic mechanism for penile erection
6. Contribution to defense mechanisms by delivering antibodies, platelets and leucocytes to affected areas of the body.

Heart



Physiologic Anatomy of Heart

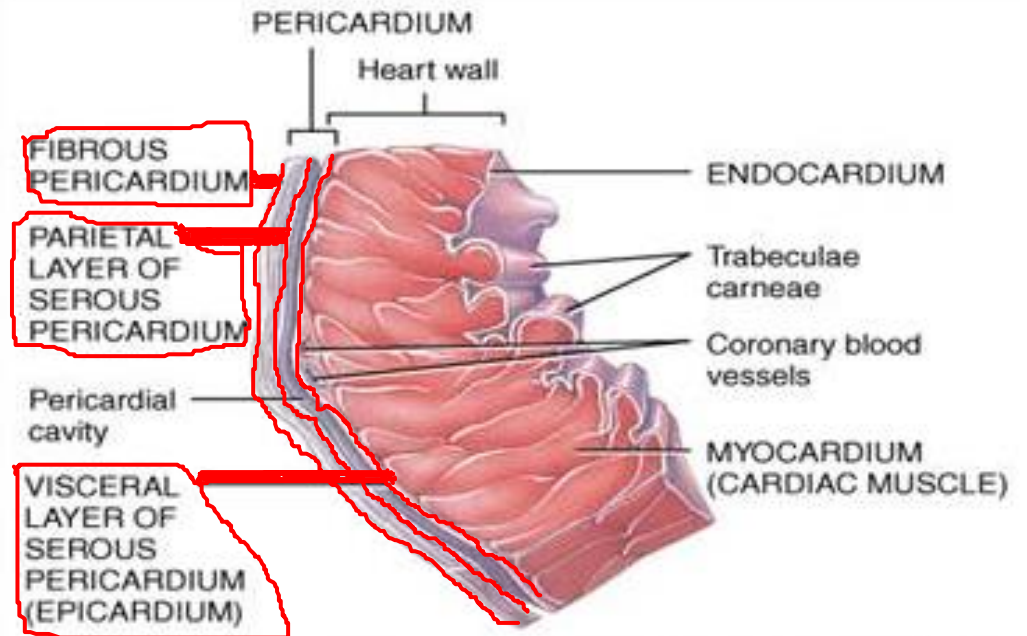
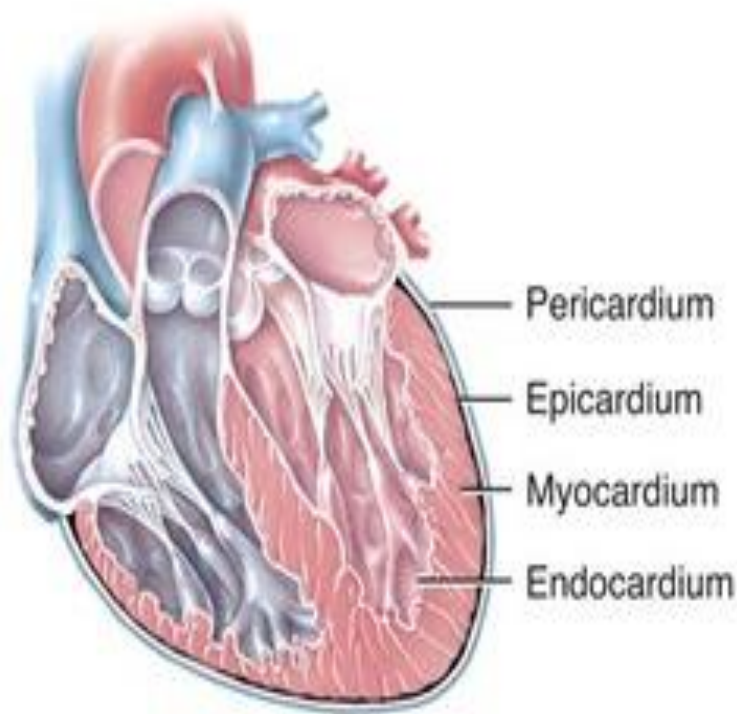
- Approximately the size of your fist
- **Weight:** 280 to 320 grams in average adults.
- Location
 - Superior surface of diaphragm
 - Left of the midline
 - Anterior to the vertebral column and posterior to the sternum



Physiologic Anatomy of Heart

The sectioned wall of the heart shows the following distinct layers from outer to deeper:

- Pericardium
- Epicardium,
- Myocardium, and
- Endocardium

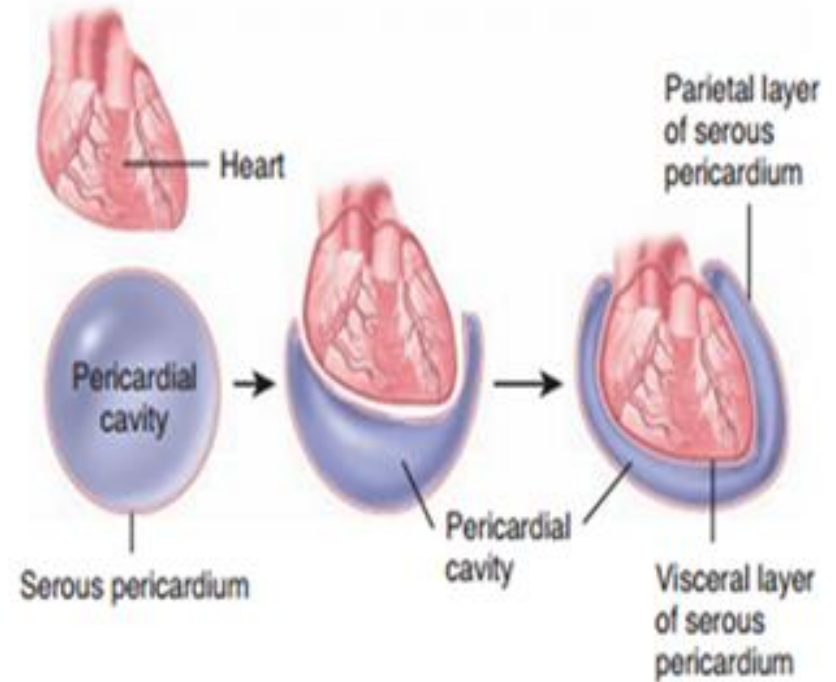
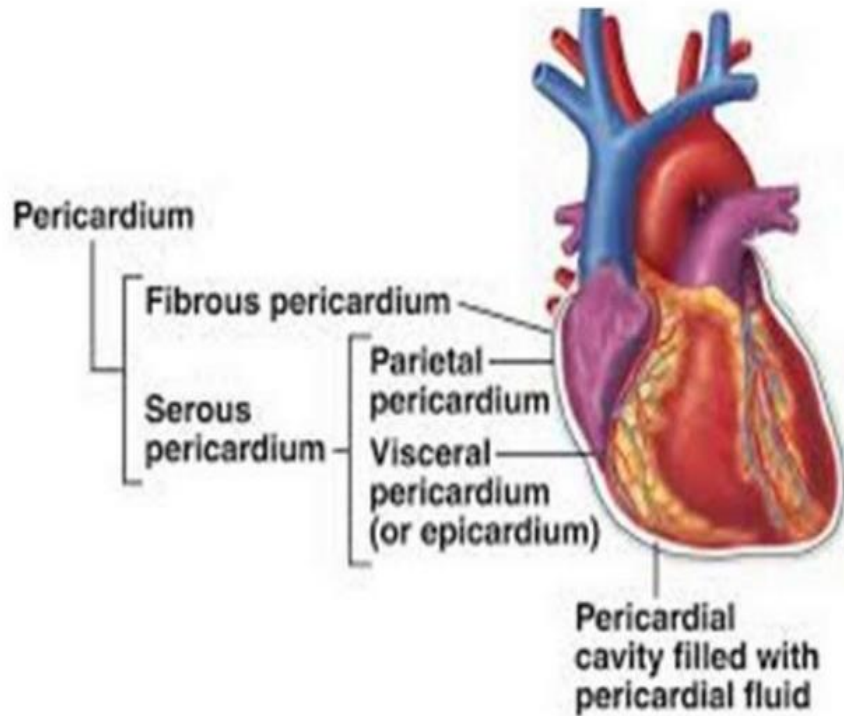


(a) Portion of pericardium and right ventricular heart wall showing the divisions of the pericardium and layers of the heart wall

Wall of the Heart

- **Pericardium** is a double-walled sac around the heart composed of:
 1. A superficial *fibrous pericardium*
 2. A deep *two-layer serous pericardium*
 - a. The parietal layer lines the internal surface of the fibrous pericardium
 - b. The visceral layer or epicardium lines the surface of the heart
 - They are separated by the fluid-filled pericardial cavity
- The Function of the Pericardium:
 - Protects and anchors the heart
 - Prevents overfilling of the heart with blood
 - Allows for the heart to work in a relatively friction-free environment

Wall of the Heart - Pericardium

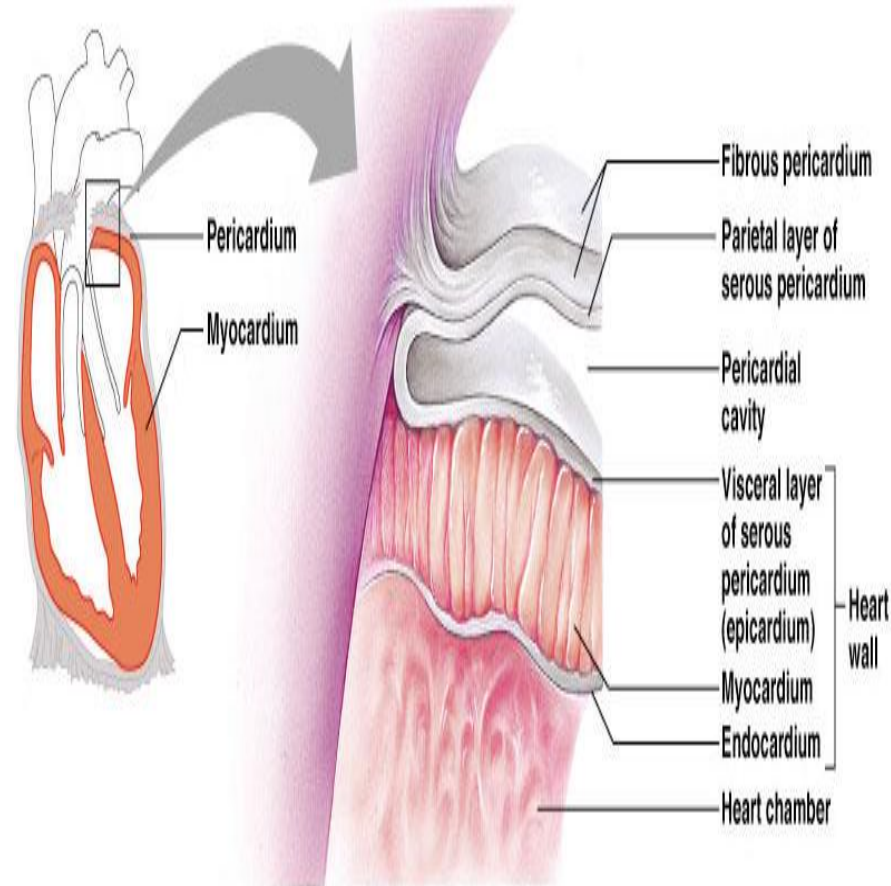


Wall of the Heart - Pericardium

- Pathogens can infect the pericardium, producing inflammation and the condition **pericarditis**.
- As a result of irritation to the pericardium, there is chest pain that may extend to the left shoulder and down the left arm.
- Treated with drugs that reduce inflammation and pain, like **aspirin**.
- **The epicardium** is the next layer in the heart wall.
- The epicardium contains **nerves, lymphatics, and blood vessels** that supply the myocardium.

Layers of the Heart

- *Epicardium* – visceral layer of the serous pericardium
- *Myocardium* – cardiac muscle layer forming the bulk of the heart
- *Fibrous skeleton of the heart* – crisscrossing, interlacing layer of connective tissue
- *Endocardium* – endothelial layer of the inner myocardial surface

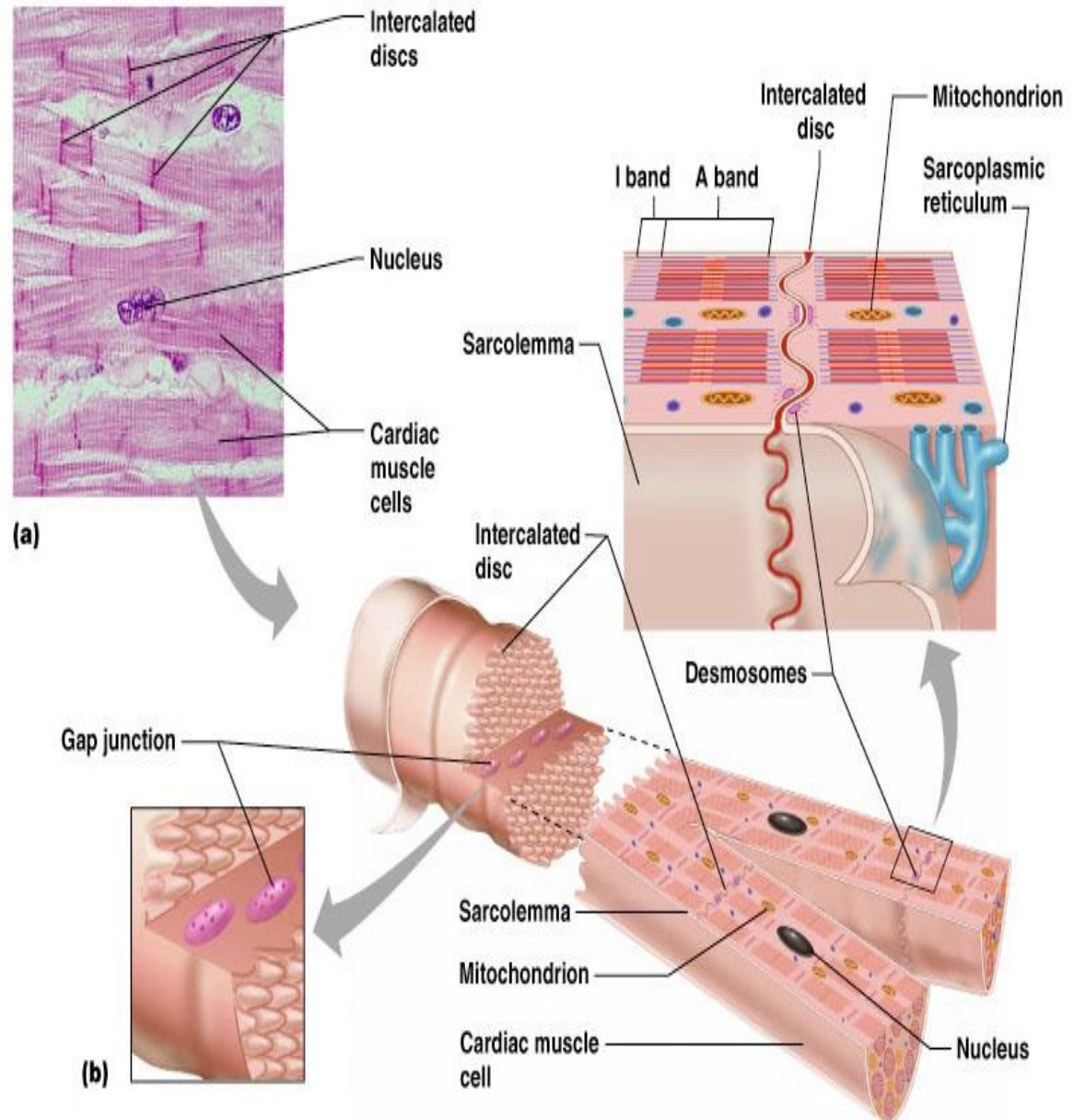


Layers of the Heart - Myocardium

- Is muscle of the heart and includes the following
 - *The atrial muscle*
 - *The ventricular muscle*
 - *The specialized excitatory and conductive muscle fibres (Autorhythmic Cells)*
- Atrial and ventricular muscles are contractile components (Contain too many actin & myosin)
- The atrial and the ventricular muscles are separated by the **fibrous skeleton of the heart**.
- Specialized excitatory muscle are weakly contractile but qualified for autorhythmicity

Layers of the Heart - Myocardium

- Cardiac muscle cells are **striated**, **mono-nucleated** and **branched**, connecting to each other through **intercalated disks** that include **gap junctions**, for chemical and electrical communication.
- Cardiac muscle has many **mitochondria** and utilizes most of the O₂

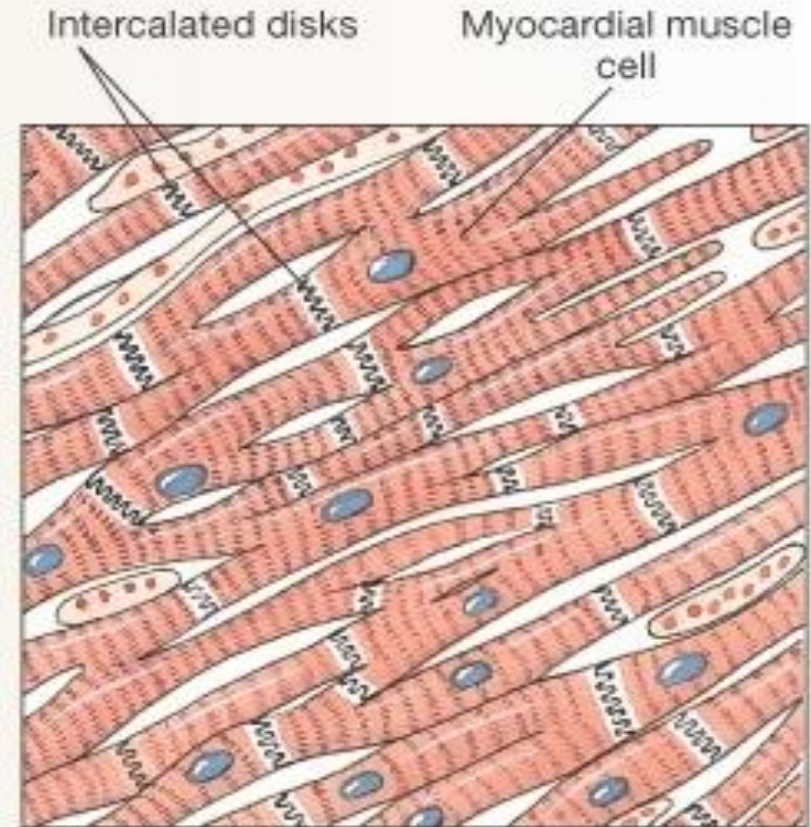


Comparison of Cardiac and Skeletal Muscle

Structural Comparison	Heart Muscle	Skeletal Muscle
Contractile proteins: actin and myosin	Present	Present
Shape of muscle fibers	Shorter than skeletal muscle fibers and branching	Elongated—no branching
Nuclei	Single	Multiple
Z discs	Present	Present
Striated	Yes	Yes
Cellular junctions	Yes—intercalated discs	No junctional complexes
Connective tissue	Endomysium	Epimysium, perimysium, and endomysium
Functional Comparison		
Energy production	Aerobic (primarily)	Aerobic and anaerobic
Calcium source (for contraction)	Sarcoplasmic reticulum and extracellular calcium	Sarcoplasmic reticulum
Neural control	Involuntary	Voluntary
Regeneration potential	None—no satellite cells present	Some possibilities via satellite cells

Layers of the Heart - Myocardium

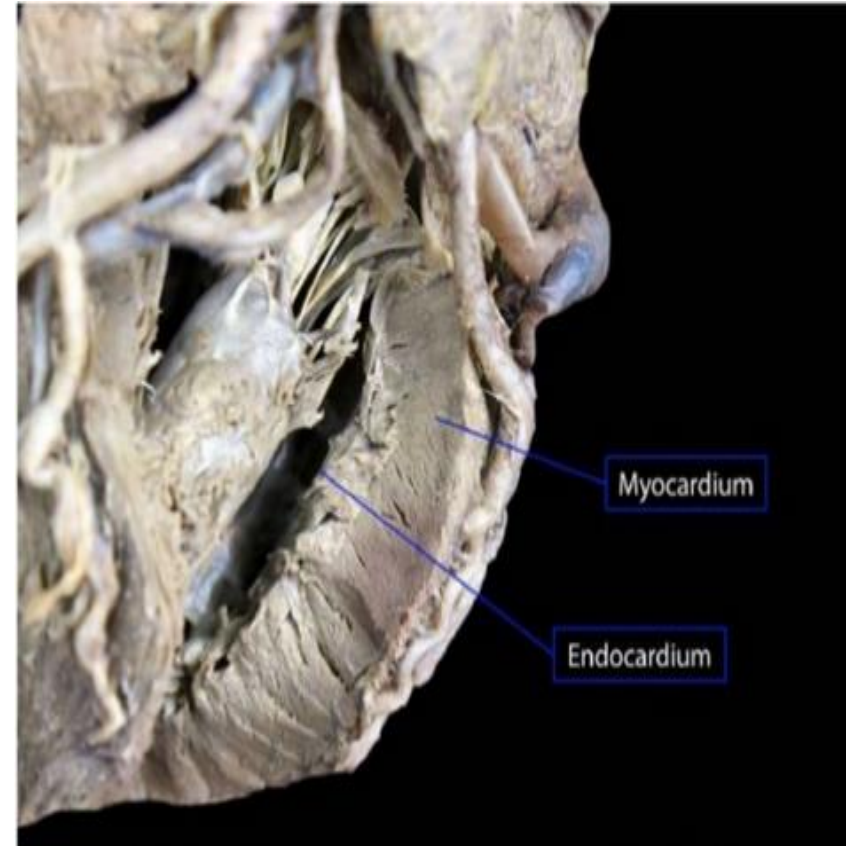
- Intercalated discs contain two types of specialized junctions:
- (a) **Desmosomes** (which act like rivets & hold the cells tightly together) and (b) **Gap junctions** (which permit action potentials to easily spread from one cardiac muscle cell to adjacent cells)
- Cardiac muscle tissue forms 2 functional syncytia or units:
(a) The atria syncytium & (b) The ventricles syncytium



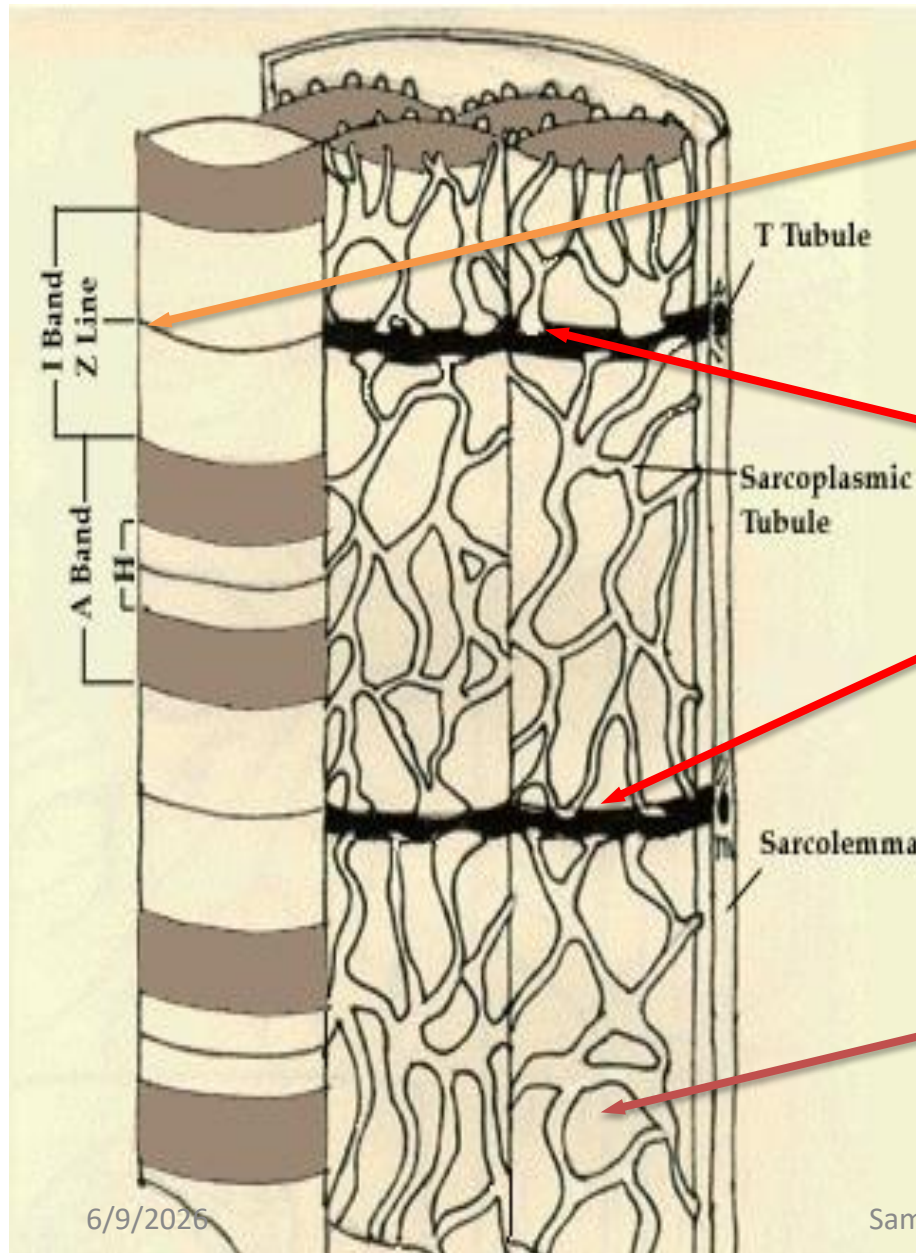
Myocardial muscle cells are branched, have a single nucleus, and are attached to each other by specialized junctions known as intercalated disks.

Layers of the Heart - Myocardium

- It has the same arrangement with skeletal muscle in actin and myosin, bands, Z-discs, zone and T-tubles.
- *T-tubles* are wider and less in amount as compared with skeletal muscles.
- As a result they posses smaller sarcoplasmic reticulum, cardiac muscle has a *smaller interacellular Ca^{+2} reserve*.



Layers of the Heart - Myocardium



T-tubules, larger than in skeletal muscle, are aligned with Z-disks (one per sarcomere).

SR is juxtaposed with T-tubules at very small terminal bulbs, rather than large cisternae.

SR is thin with much smaller volume than skeletal muscle.

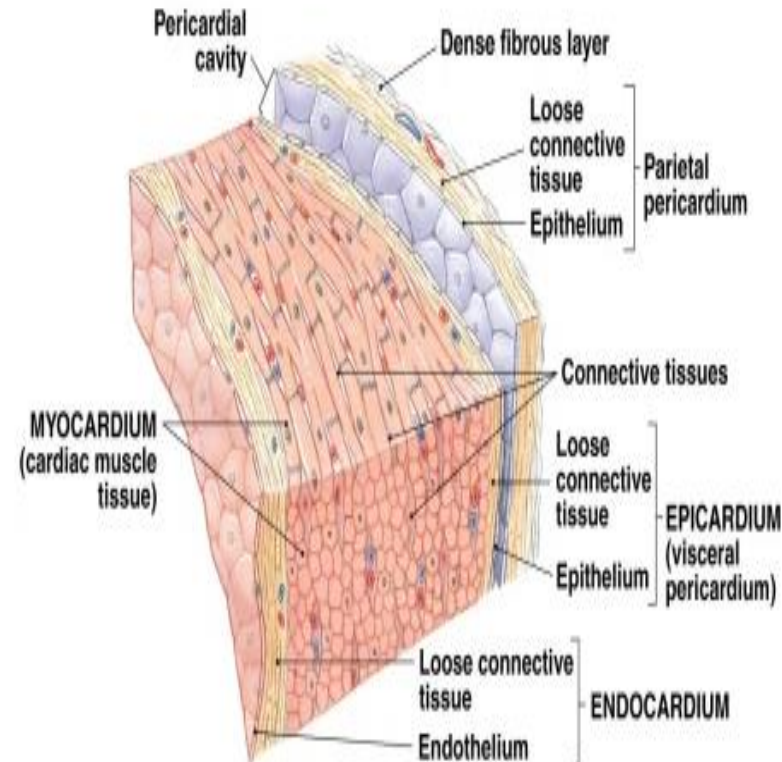
Layers of the Heart - Endocardium

- The innermost endocardium is a thin layer of **endothelium** overlying a thin layer of connective tissue.
- It provides a smooth lining for the chambers of the heart and covers the valves of the heart.
- The smooth endothelial lining minimizes the surface friction as blood passes through the heart.
- The endocardium is continuous with the endothelial lining of the large blood vessels attached to the heart.

Layers of the Heart - Endocardium

Endocardium

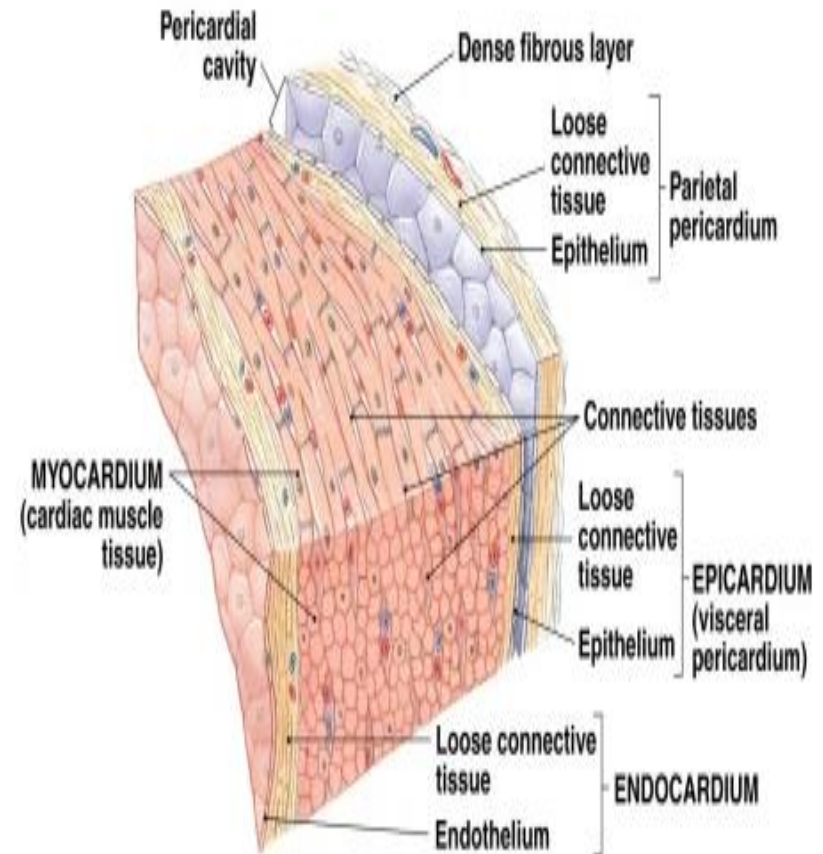
- The innermost layer of the cardiac wall is known as the endocardium. It lines the **cavities** and **valves** of the heart.
- Structurally, the endocardium is comprised of loose connective tissue and simple squamous epithelial tissue
 - it is similar in its composition to the endothelium which lines the inside of blood vessels.
- In addition to lining the inside of the heart, the endocardium also **regulates contractions** and aids cardiac **embryological development**



Layers of the Heart - Endocardium

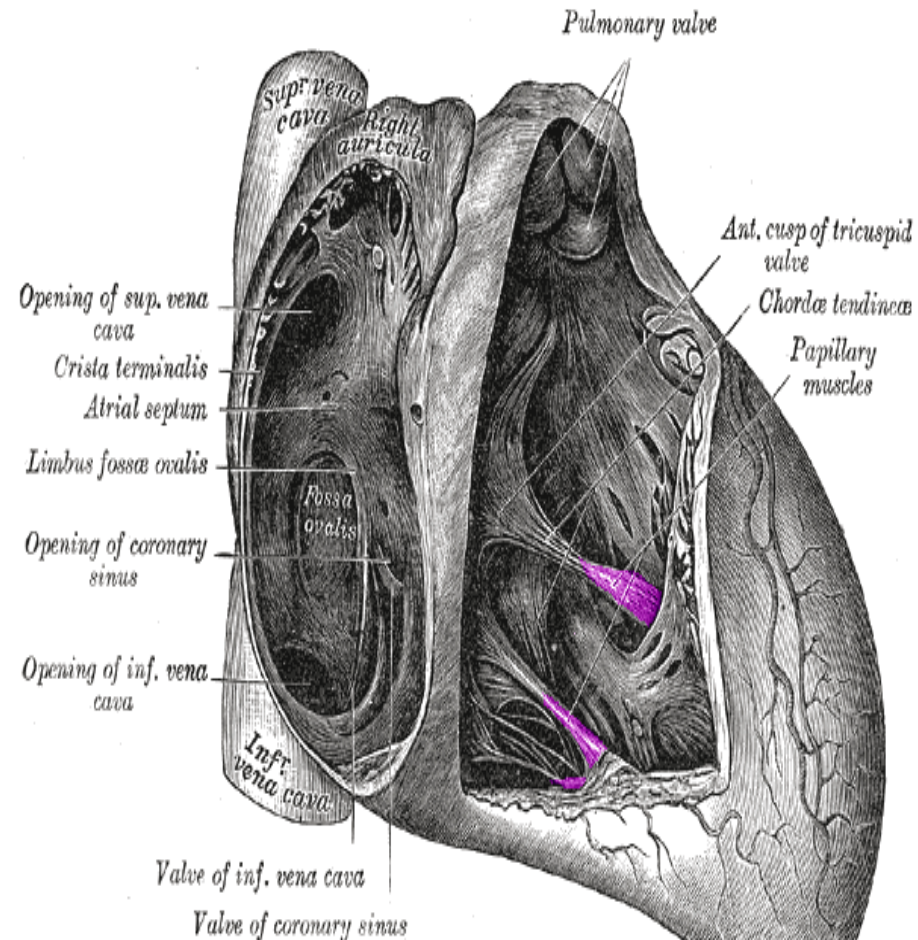
Subendocardial layer

- The subendocardial layer lies between, and joins, the **endocardium** and the **myocardium**.
- It consists of a layer of loose fibrous tissue, containing the vessels and nerves of the conducting system of the heart. The **purkinje fibres** are located in this layer.
- As the subendocardial layer houses the conducting system of the heart, damage to this layer can result in various **arrhythmias**.

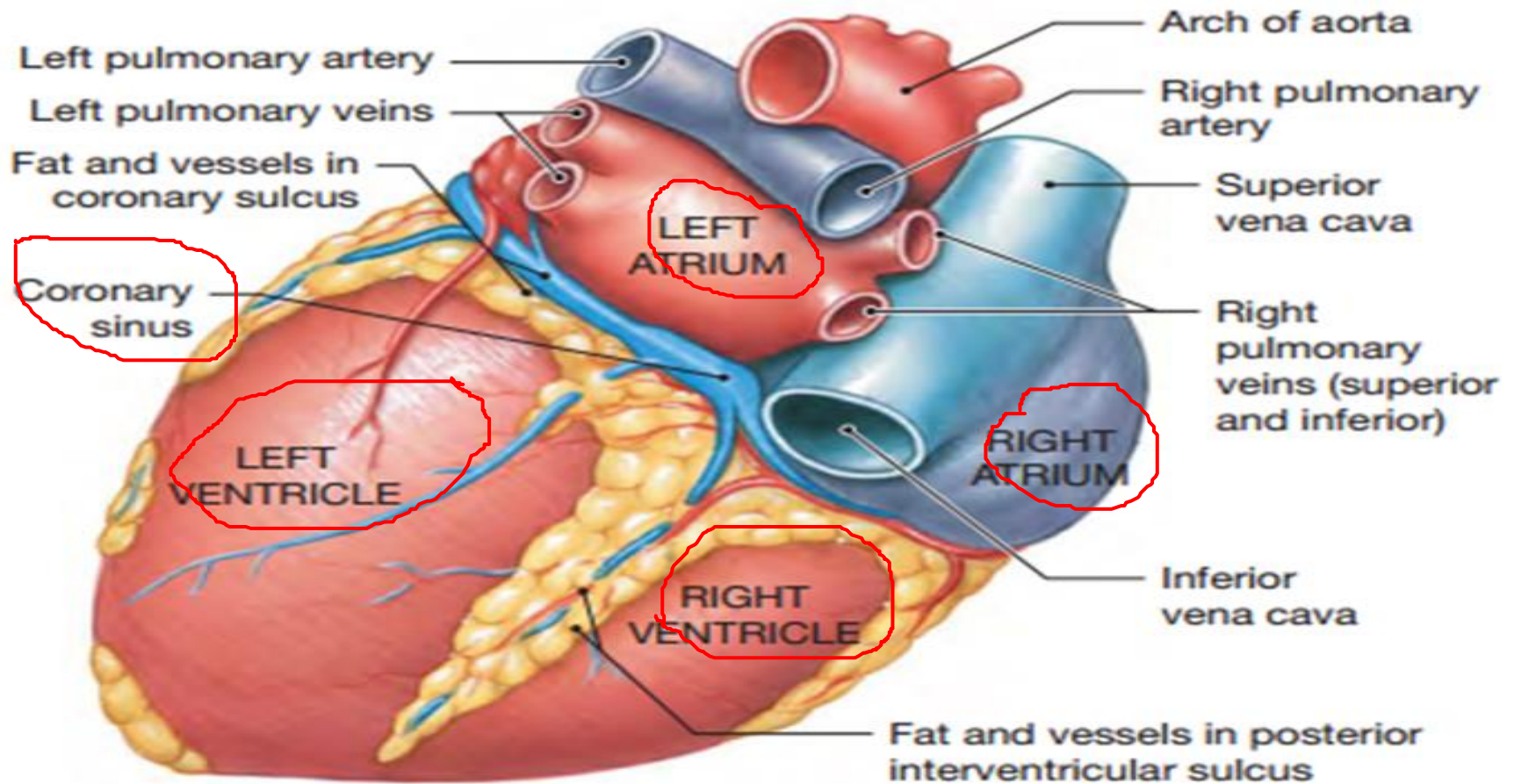


Physiologic Anatomy of Heart

- The heart has four chambers.
- The two superior receiving chambers *are the atria*, and the two inferior pumping *chambers are the ventricles*.
- The paired atria receive blood from blood vessels returning blood to the heart, called veins, while the ventricles eject the blood from the heart into blood vessels called arteries.
- A deep groove, marks the border between the atria and the ventricles called *as coronary sulcu*.



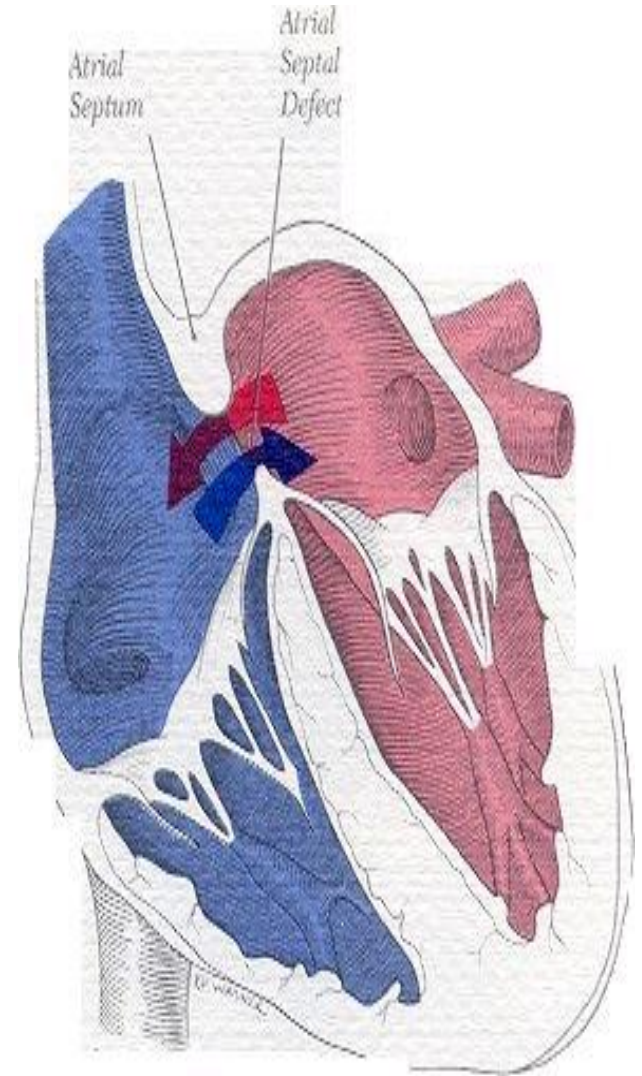
Physiologic Anatomy of Heart



b Major landmarks on the posterior surface. Coronary arteries (which supply the heart itself) are shown in red; coronary veins are shown in blue.

Physiologic Anatomy of Heart

- On the interior surface of the heart the right atrium communicates with the right ventricle, and the left atrium with the left ventricle.
 - The atria are separated by the **inter-atrial septum**.
 - The ventricles are separated by the much thicker **inter-ventricular septum**.
- A prominent feature of inter-atrial septum is an oval depression called the **fossa ovalis**, the remnant of the **foramen ovale**.
- It is an opening in the interatrial septum of the fetal heart that normally closes soon after birth.

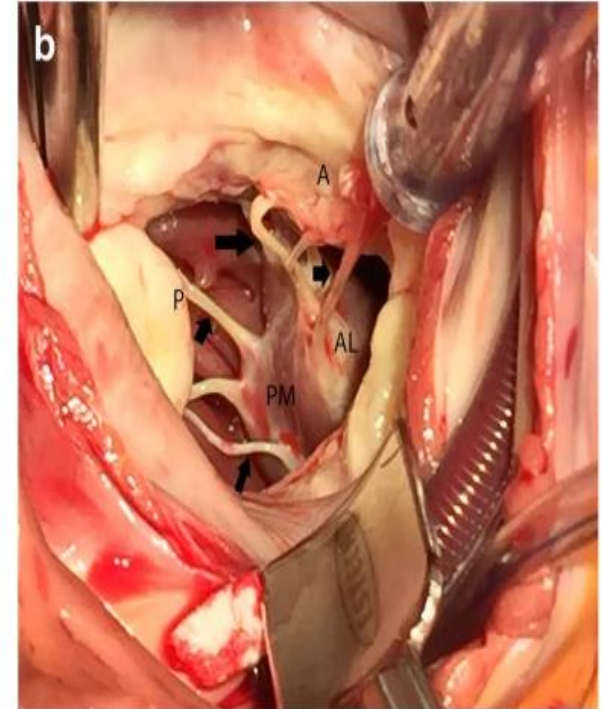
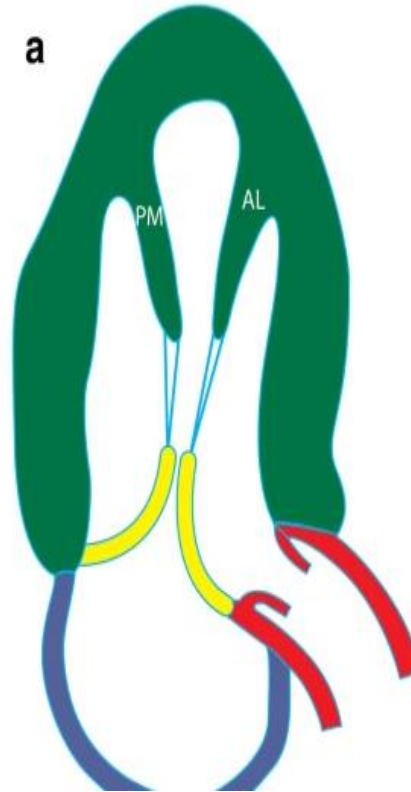


Physiologic Anatomy of Heart

- Blood passes from one side to the other side of the heart by the help of valves.
 - The heart valves are lined with endocardium.
- Heart got two major types of valves:
 1. The **atrioventricular** (bi-cuspid /mitral and tri-cuspid) valves prevent the backflow of blood from the ventricles to the atria when the ventricles are contracting.
 2. The **Semilunar valves** (pulmonary and aortic valves) prevent the backflow of blood from the pulmonary trunk and aorta into the right and left ventricles, respectively.

Papillary Muscles

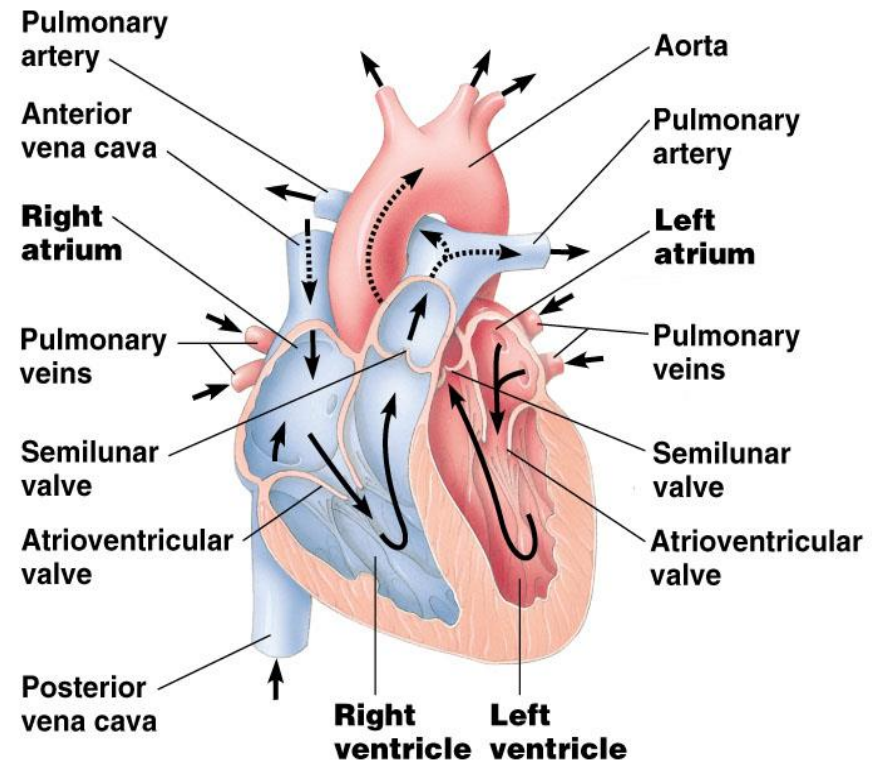
- The papillary muscles originate from the free wall of the LV attached to trabecula carnea.
- The papillary muscles give rise to multiple chordae tendinae (blue), which attach to the mitral valve (yellow).



- *Illustration showing the anterolateral (AL) and posteromedial (PM) projection of mitral papillary muscles*

The Cardiac Valves

- Cardiac valves are made up of **fibrous connective tissues**.
 - *There are 4 cardiac valves:*
 - 1. Tricuspid valve (T-valve)**
 - 2. Bicuspid valve (M-valve)**
 - 3. Pulmonary semilunar valve (P-valve)**
 - 4. Aortic semilunar valve (A-valve)**
- ❖ **Function:** prevent back flow of blood.

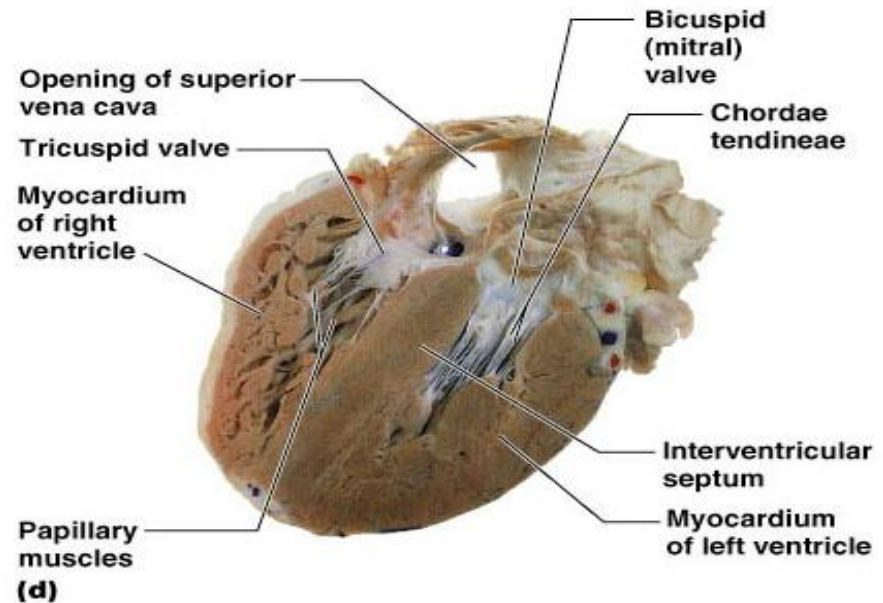
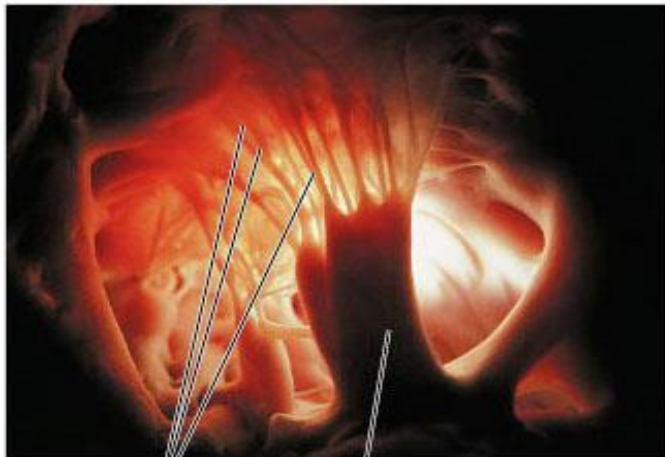
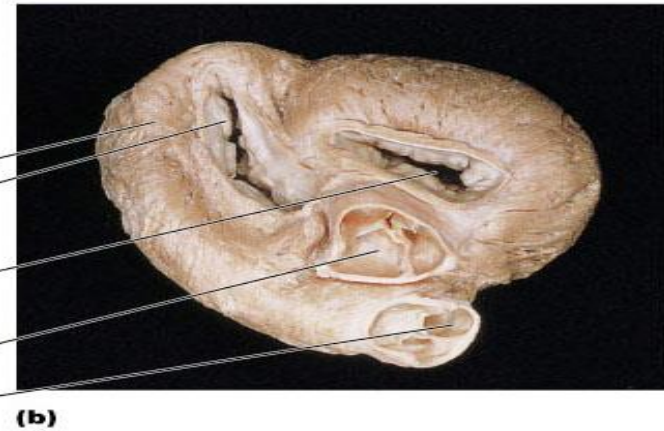
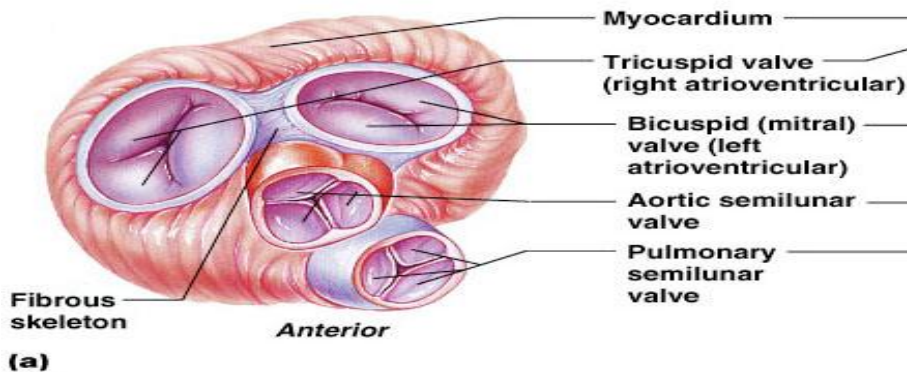
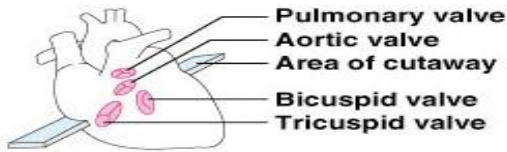


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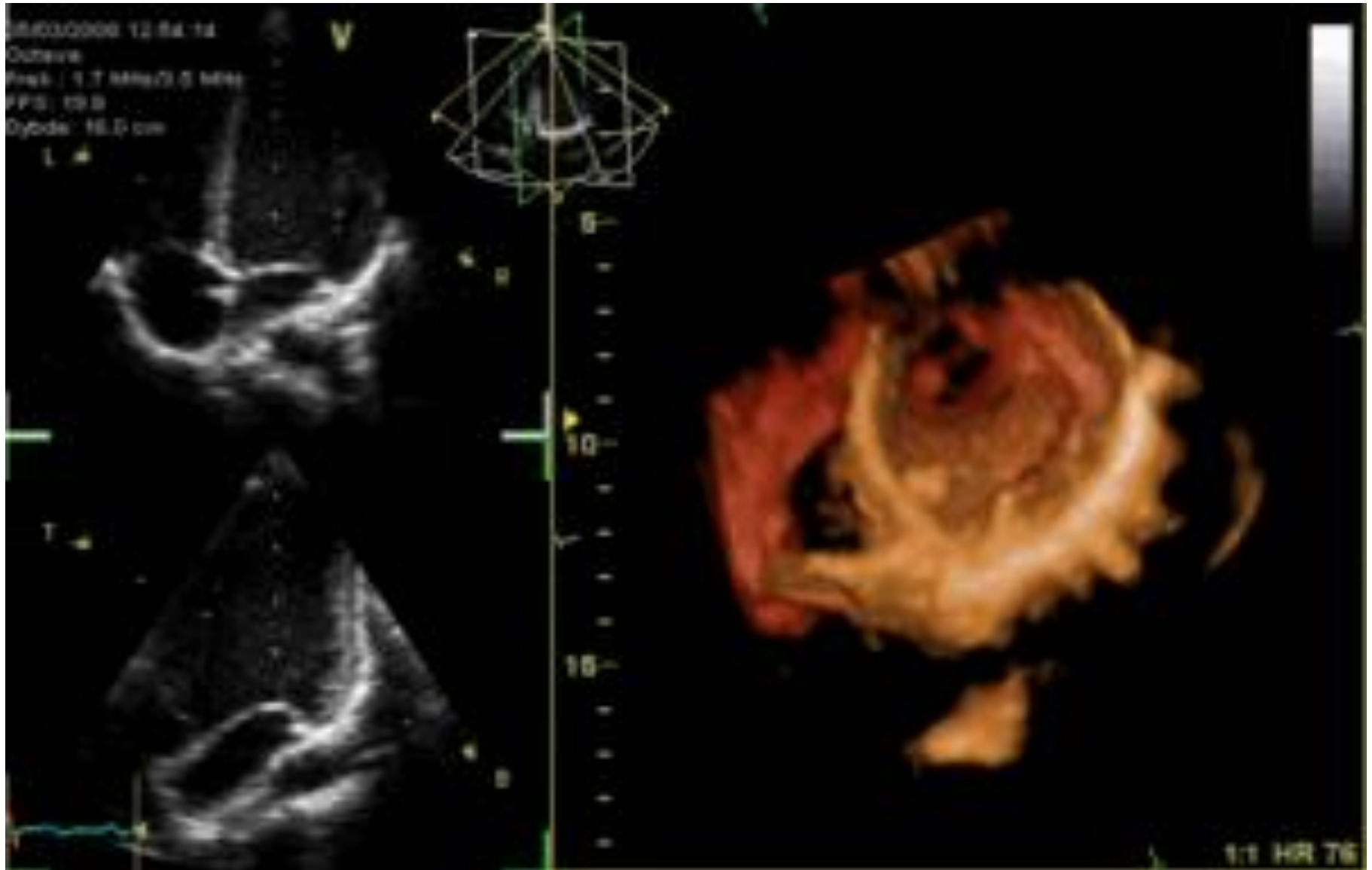
Cardiac Valves, cont'd

- **Atrioventricular valves** - are located b/n atria & ventricles.
- **Aortic & pulmonary valves**-are known as semilunar valves because they are made up of 3 crescent moon-shaped cusps.
- If the *AV valves are damaged*, the blood may *regurgitate* (backflow) into atria when ventricles contract.
- ⊕ All valves consist of *connective tissue* (not a cardiac muscle tissue) and, therefore, *open & close passively*.
- ⊕ Valves open & close in response to *changes in pressure*:

Heart Valves

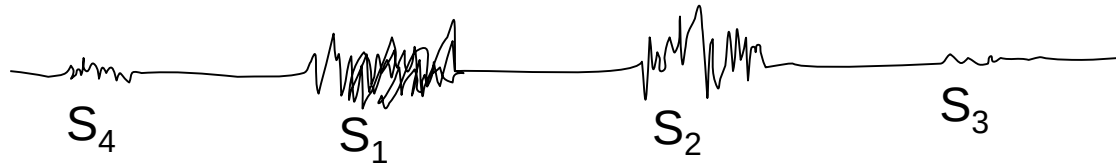


Types of Cardiac valves...cont'd



Heart Sounds

- Heart sounds (lub-dub) are associated with **closing** of heart valves.
- 4-separate audible heart sounds (S_1 , S_2 , S_3 and S_4)
- Means of identification
 1. Auscultation: - direct/immediate auscultation
- Stethoscope mediated auscultation
 2. Phonocardiography based recording



S_1 : is always audible, has a LUB-sound

S_2 : is always audible, has DUB-sound

Continuous heart sound: Lub-Dub, Lub-Dub

Duration of Lub is shorter than that of Dub

S_3 : is audible in children & in adults during exercise

S_4 : is audible very rarely.

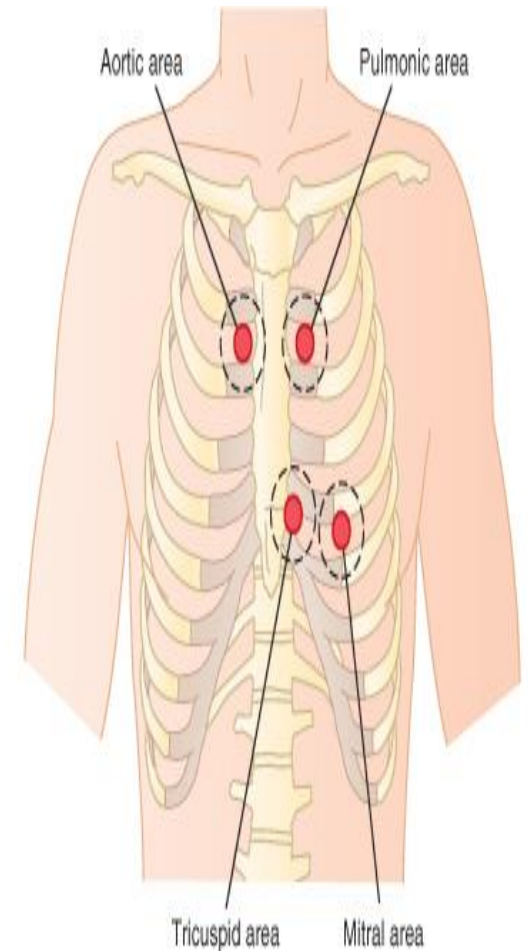
Heart Sounds (cont'd)

S₁: First heart sound

- **Cause:** sudden closure of AV-valves.
- **Timing:** occurs at the beginning of ventricular systole
- **Best site of auscultation**
 - ❖ 5th LIS (left intercostal space), at mid-clavicular line
 - ❖ 4th LIS, near the sternum

S₂: Second heart sound

- **Cause:** Sudden closure of semilunar valves
- **Timing:** occurs at the beginning of ventricular diastole
- **Best site of auscultation**
 - ❖ 2nd RIS, near the sternum
 - ❖ 2nd LIS, near the sternum



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Heart Sounds (cont'd)

S₃: Third heart sound

- **Cause:** rapid filling of the ventricles with blood during **ventricular diastole**.
 - It is audible in **children** and in adults during **exercise**.
- **Best site of auscultation:** at the **apex of the heart** beneath the xiphoid process.

S₄: Fourth heart sound

- **Cause:** rapid ventricular filling during **atrial systole**
- **Timing:** during **atrial systole**
- **Best site of auscultation:** at the apex of the heart



Abnormal Heart Sounds

- ❑ **Abnormal heart sounds** are most commonly recorded when the heart valves are defective.
- ❑ Abnormal heart sounds are called **heart murmurs**.
- ❑ They are **two types**:
 - ⊕ **Systolic murmurs**: - AV-valve insufficiency
 - Semilunar valve stenosis
 - High CO (physiological)
 - ⊕ **Diastolic murmurs**:- Semilunar valve insufficiency
 - AV-valve stenosis
- Ⓢ **Valvular insufficiency** causes **regurgitation**.